

Roll No.

Total No. of Questions : 9] [Total No. of Printed Pages : 4
(1049)

**BCA (CBCS) RUSA IInd Semester
Examination**

4385

MATHEMATICS-II

Paper : BCA-0201

Time : 3 Hours]

[Maximum Marks : 70

Note :- Attempt *one* question from each Unit. Q. No. 9 is compulsory.

Unit-I

1. (a) Show that between any two roots of $e^x \cos x = 1$, there exists at least one root of $e^x \sin x - 1 = 0$.

- (b) Verify Lagrange's mean value theorem for the function :

$f(x) = x(x-1)(x-2)$ in $\left[0, \frac{1}{2}\right]$ 5.5

2. (a) Show that :

$$\frac{\sin \alpha - \sin \beta}{\cos \beta - \cos \alpha} = \cot \theta$$

Where $0 < \alpha < \theta < \beta < \frac{\pi}{2}$.

CH-711

(1)

Turn Over

(b) State and prove Leibnitz theorem.

5,5

Unit-II

3. (a) Show that for any two integers a and $b > 0$, there exists q_1 and r_1 such that

$$a = bq_1 + cr_1, 0 \leq r_1 < \frac{b}{2}, c = 1 \text{ or } -1.$$

- (b) Find the remainder when 2^{340} is divided by 341.

$$2^{340} \div 341 = 15 \cdot 20 \cdot 25$$

5,5

4. (a) Solve $12x + 15 \equiv 0 \pmod{45}$.

- (b) If p is prime, show that $2(p-3)! + 1$ is a multiple of p .

5,5

Unit-III

5. (a) Show that $S = \{3^n : n \in \mathbb{Z}\}$ is a commutative group with respect to multiplication.

$$\begin{array}{r} 12 \cdot 45 \cdot 7 \\ - 24 \cdot 1 \\ \hline 4 \cdot 10 \cdot 1 \\ - 3 \cdot 10 \cdot 1 \\ \hline 1 \end{array}$$

- (b) Prove that the set of matrices :

$$G = \left\{ \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} : \alpha \in \mathbb{R} \right\}$$

forms a group under matrix multiplication.

5,5

6. (a) Show that a finite semi-group G in which cancellation laws hold is a group.

- (b) Prove that the group G of n n^{th} roots of unity is a cyclic group. 5,5

Unit-IV

7. (a) Prove that a ring R is commutative if and only if

$$(a + b)^2 = a^2 + 2ab + b^2 \quad \forall a, b \in R$$

- (b) Show that the ring $\mathbb{Z}_p$ of integers modulo p is a field if and only if p is prime. 5,5

8. (a) Let $f(x)$ and $g(x)$ be two non-zero polynomials in $R[x]$, R being a ring. If $f(x) + g(x) \neq 0$, then prove that :

$$\deg(f(x) + g(x)) \leq \max(\deg f(x), \deg g(x))$$

- (b) Find the sum and product of the polynomials :

$$f(x) = 4x - 5, g(x) = 2x^2 - 4x + 2 \text{ in } \mathbb{Z}_8[x] \quad 5,5$$

Unit-V

(Compulsory Question)

9. (a) State Lagrange's mean value theorem.

(b) $\lim_{x \rightarrow \infty} \frac{\sin x}{x} = \dots\dots\dots$

- (c) Find 100th derivative of x^{101} .

- (d) Find the remainder when 3^{287} is divided by 23.
- (e) L.C.M. (8, 12, 15, 20, 25) =
- (f) Give an example of an infinite abelian group.
- (g) The set $\mathbb{Z}$ of all integers is an abelian group under the binary composition $a*b = a + b + 2$. Find the identity element.
- (h) Prove that non-abelian groups cannot be cyclic.
- (i) Define ring and ring with unity.
- (j) Prove that $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$, the ring of integer modulo 6, is with zero divisors. $3 \times 10 = 30$